AWS Simple Storage Service S3

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Amazon Simple Storage Service (S3) provides object storage through a web interface.

S3 provides high availability, durability, and cost-effective storage for backups, big data analytics, and cloud applications.

S3 Bucket

- A bucket in Amazon S3 is a container for storing objects (files).
- Buckets are created in a specific AWS region and have globally unique names.
- Buckets are defined at the region level.
- You can control access to buckets using bucket policies, ACLs, and IAM roles.

Example:

- Bucket: my-bucket-unique

S3 Bucket

Amazon S3 > Buckets > Create bucket

Create bucket Info

Buckets are containers for data stored in S3.

General configuration

AWS Region

Europe (Frankfurt) eu-central-1

Bucket name | Info

my-aws-bucket-unique

Bucket name must be unique within the global namespace and follow the bucket naming rules. See rules for bucket naming

Copy settings from existing bucket - *optional*

Only the bucket settings in the following configuration are copied.

Choose bucket

Format: s3://bucket/prefix

Object Ownership Info

Control ownership of objects written to this bucket from other AWS accounts and the use of access control lists (ACLs). Object ownership determines who can specify access to objects.

- (●) ACLs disabled (recommended)
  All objects in this bucket are owned by this account. Access to this bucket and its objects is specified using only policies.
- ( ) ACLs enabled
  Objects in this bucket can be owned by other AWS accounts. Access to this bucket and its objects can be specified using ACLs.

Object Ownership

Bucket owner enforced

S3 Objects (Files)

- Files stored in S3 are called objects.
- Each object consists of data, a key (file name, full path), metadata, and optionally tags.
- Objects are stored within buckets and identified using a unique key (like a file path).

Example:

- Bucket: my-bucket-unique
- Key: AWS-logo-s3.png
- S3 address:
  - s3://my-bucket-unique/AWS-logo-s3.png

S3 Features

- Max. Object Size is 5TB (5000GB)
- If uploading more than 5GB, it must be used multi-part upload
- Scalability: Unlimited storage.
- Durability: 99.999999999% (11 9s) durability.
- Security: Encryption and access management.
- Cost-effectiveness: Multiple storage classes.

How to create S3 Bucket?

Amazon S3 > Buckets > Create bucket

Create bucket Info

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How to create S3 Bucket?

Block Public Access settings for this bucket

Public access is granted to buckets and objects through access control lists (ACLs), bucket policies, access point policies, or all. In order to ensure that public access to this bucket and its objects is blocked, turn on Block all public access. These settings apply only to this bucket and its access points. AWS recommends that you turn on Block all public access, but before applying any of these settings, ensure that your applications will work correctly without public access. If you require some level of public access to this bucket or objects within, you can customize the individual settings below to suit your specific storage use cases. Learn more

- [x] **Block *all* public access**
  Turning this setting on is the same as turning on all four settings below. Each of the following settings are independent of one another.
  - [x] **Block public access to buckets and objects granted through *new* access control lists (ACLs)**
    S3 will block public access permissions applied to newly added buckets or objects, and prevent the creation of new public access ACLs for existing buckets and objects. This setting doesn't change any existing permissions that allow public access to S3 resources using ACLs.
  - [x] **Block public access to buckets and objects granted through *any* access control lists (ACLs)**
    S3 will ignore all ACLs that grant public access to buckets and objects.
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  - [x] **Block public and cross-account access to buckets and objects through *any* public bucket or access point policies**
    S3 will ignore public and cross-account access for buckets or access points with policies that grant public access to buckets and objects.

Bucket Versioning

Versioning is a means of keeping multiple variants of an object in the same bucket. You can use versioning to preserve, retrieve, and restore every version of every object stored in your Amazon S3 bucket. With versioning, you can easily recover from both unintended user actions and application failures. Learn more

Bucket Versioning

- (•) Disable
- ( ) Enable

How to create S3 Bucket?

Default encryption Info

Server-side encryption is automatically applied to new objects stored in this bucket.

Encryption type | Info

- (●) Server-side encryption with Amazon S3 managed keys (SSE-S3)
- ( ) Server-side encryption with AWS Key Management Service keys (SSE-KMS)
- ( ) Dual-layer server-side encryption with AWS Key Management Service keys (DSSE-KMS)
  Secure your objects with two separate layers of encryption. For details on pricing, see **DSSE-KMS pricing** on the **Storage** tab of the Amazon S3 pricing page.

Bucket Key

Using an S3 Bucket Key for SSE-KMS reduces encryption costs by lowering calls to AWS KMS. S3 Bucket Keys aren't supported for DSSE-KMS. Learn more

- ( ) Disable
- (●) Enable

▼ Advanced settings

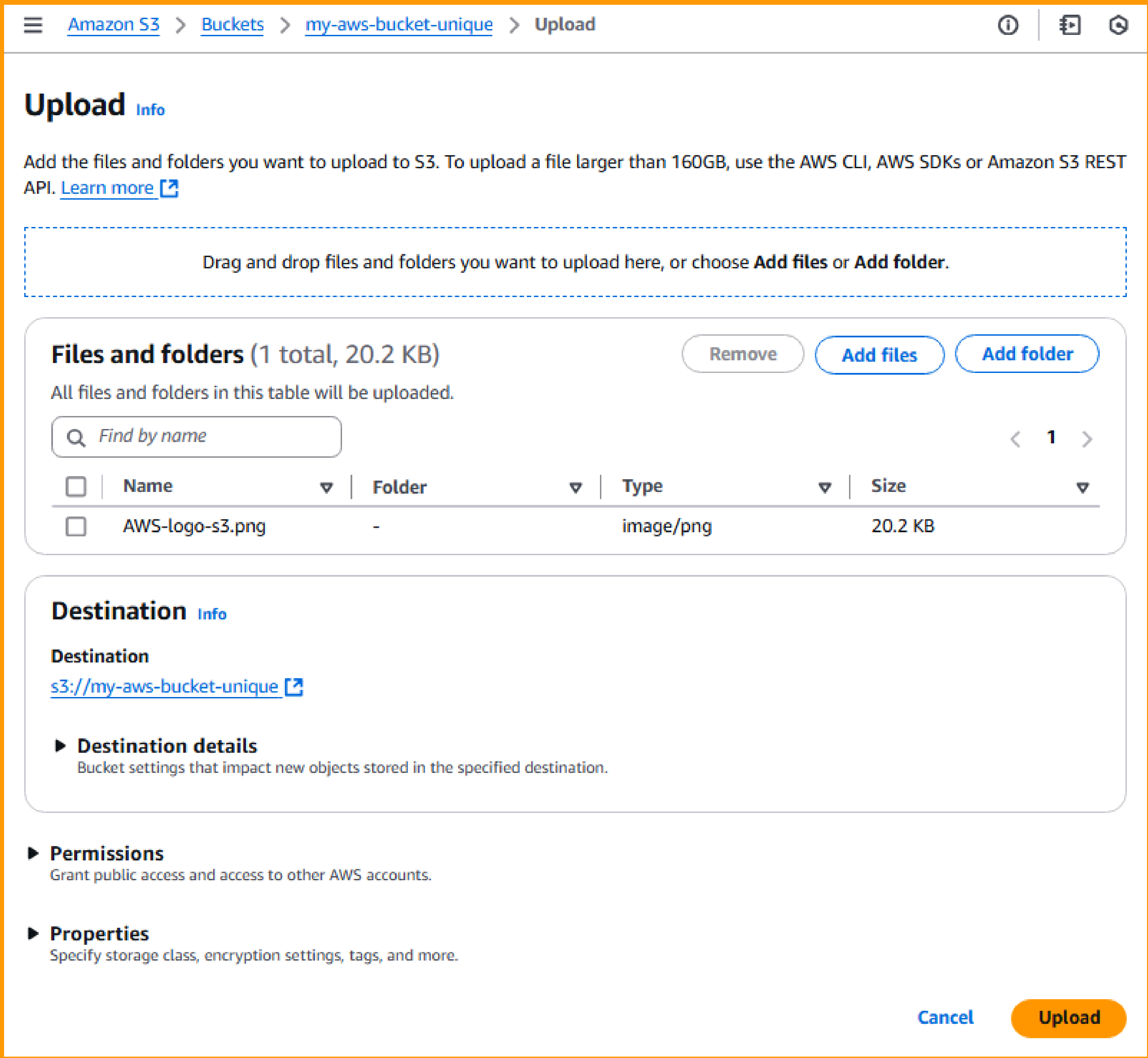
Object Lock

Store objects using a write-once-read-many (WORM) model to help you prevent objects from being deleted or overwritten for a fixed amount of time or indefinitely. Object Lock works only in versioned buckets. Learn more

- (●) Disable
- ( ) Enable
  Permanently allows objects in this bucket to be locked. Additional Object Lock configuration is required in bucket details after bucket creation to protect objects in this bucket from being deleted or overwritten.

ⓘ Object Lock works only in versioned buckets. Enabling Object Lock automatically enables Versioning.

How to create S3 Bucket?



Storage Classes

Amazon S3 offers storage classes to optimize costs based on access patterns and performance requirements.

- S3 Standard: For frequently accessed data.
- S3 Intelligent-Tiering: Automatically moves data to the most cost-efficient tier.
- S3 Glacier Instant Retrieval: Low-cost storage for infrequent access.
- S3 Glacier Flexible Retrieval: Archive data with retrieval times of minutes to hours.
- S3 Glacier Deep Archive: Lowest cost for long-term archival.

Storage Classes
S3 Lifecycle Rules

- Automatically manage object storage over time to optimize costs.
- How: Define rules to transition objects between storage classes or delete them after a set period.
- Examples:
  - Move objects from S3 Standard to S3 Glacier after 30 days.
  - Permanently delete objects older than 365 days.

S3 Event Notifications

- Trigger actions when specific events happen in your S3 bucket.
- Supported Destinations:
    - Amazon SNS: Send notifications.
    - Amazon SQS: Send messages to a queue.
    - AWS Lambda: Trigger serverless functions.
- Examples:
    - Trigger a Lambda function when a new file is uploaded.
    - Notify a monitoring system when files are deleted.

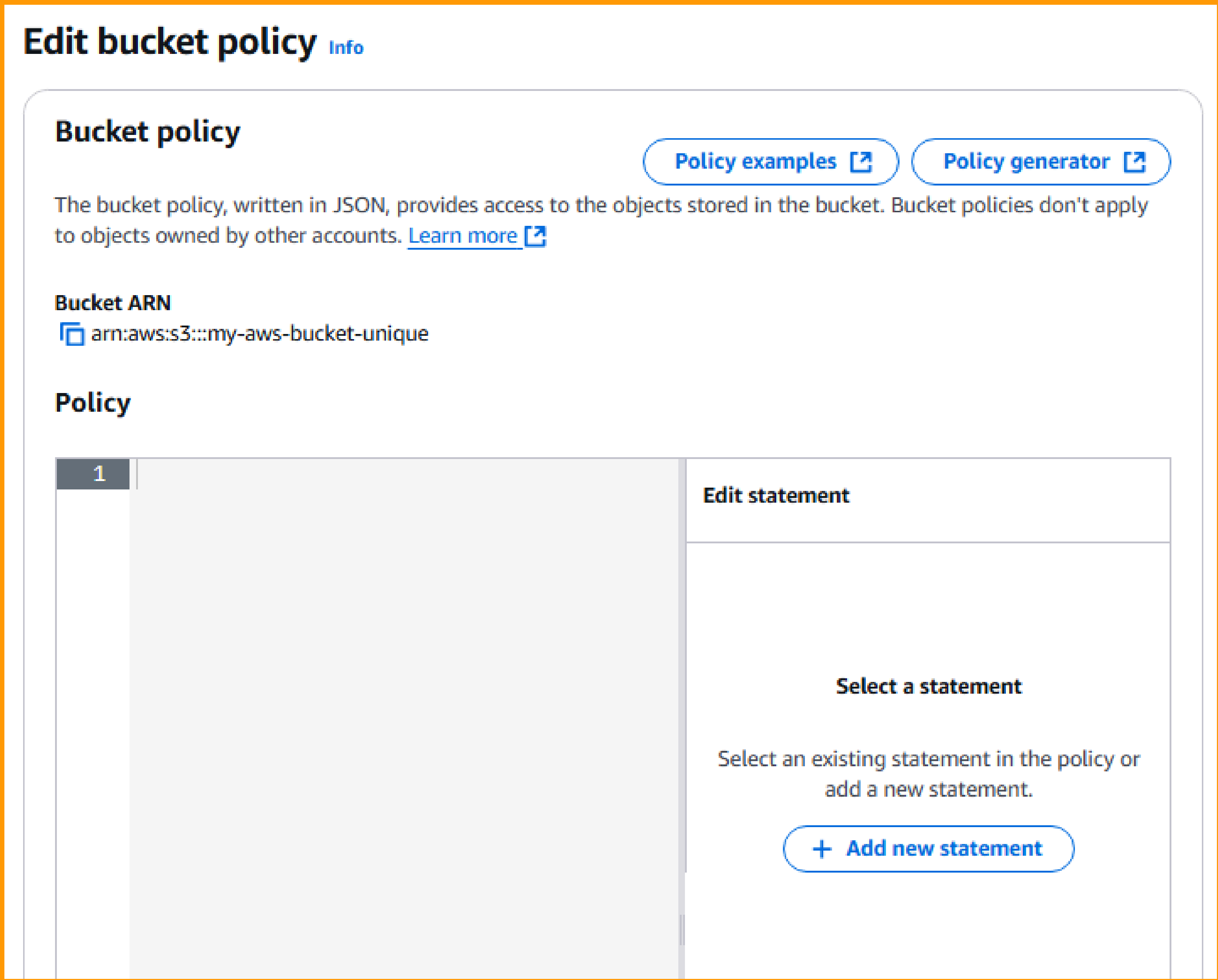
S3 Bucket Policy

- Define access permissions at the bucket level using JSON-based policies.
- **Granularity**: Controls access for IAM users, roles, or external accounts.
- **Principle of Least Privilege**: Grant the minimum necessary permissions.
- **Avoid Public Buckets**: Only make public when absolutely necessary, and monitor with AWS Trusted Advisor.
- **Use IAM Roles for Applications**: Assign IAM roles to applications instead of hardcoding credentials.
- **Enable Bucket Policy Logging**: Use AWS CloudTrail to track changes and access patterns.

S3 Bucket Policy Details

- **Version**: Defines the policy language version (always use "2012-10-17").
- **Statement**: Array of rules for access control. Each statement has:
- **Effect**: Allow or Deny.
- **Principal**: Who can access the bucket (IAM roles, users, AWS services).
- **Action**: S3 actions like s3:GetObject or s3:PutObject.
- **Resource**: The bucket or objects being accessed.
- **Condition (optional)**: Adds context-based rules like IP address restrictions.

S3 Bucket Policy Details



S3 Bucket Policy Details

AWS Policy Generator

The AWS Policy Generator is a tool that enables you to create policies that control access to Amazon Web Services (AWS) products and resources. For more information about creating policies, see key concepts in Using AWS Identity and Access Management. Here are sample policies.

Step 1: Select Policy Type

A Policy is a container for permissions. The different types of policies you can create are an IAM Policy, an S3 Bucket Policy, an SNS Topic Policy, a VPC Endpoint Policy, and an SQS Queue Policy.

Select Type of Policy S3 Bucket Policy

Step 2: Add Statement(s)

A statement is the formal description of a single permission. See a description of elements that you can use in statements.

Effect Allow Deny

Principal

Use a comma to separate multiple values.

AWS Service Amazon S3 All Services ('*')

Use multiple statements to add permissions for more than one service.

Actions -- Select Actions -- All Actions ('*')

Amazon Resource Name (ARN)

ARN should follow the following format: arn:aws:s3:::${BucketName}/${KeyName}.
Use a comma to separate multiple values.

Add Conditions (Optional)

Add Statement

Step 3: Generate Policy

A *policy* is a document (written in the Access Policy Language) that acts as a container for one or more statements.

Add one or more statements above to generate a policy.

S3 Bucket Policy
IAM Policy

- S3 Bucket Policy:
  - Add a bucket policy to explicitly allow the IAM user's principal (ARN) to access the bucket.
  - This is crucial when you need to provide access across accounts or add additional restrictions (e.g., IP conditions).
- IAM Policy (User or Role):
  - Attach an IAM policy to the user (or to an IAM role the user assumes) that grants permissions to interact with the bucket.
  - This policy defines what actions the user can perform and on which bucket.

S3 Bucket Policy

{} bucket-policy.json

{} bucket-policy.json > ...

```json
{
 "Version": "2012-10-17",
 "Statement": [
   {
     "Sid": "AllowListBucket",
     "Effect": "Allow",
     "Principal": {
          "AWS": "arn:aws:iam::<AccountID>:user/<IAMUser>"
      },
      "Action": "s3:GetObject",
      "Resource": "arn:aws:s3:::<BucketName>/*"
   }
 ]
}
```

IAM Policy

{} iam-policy.json ×

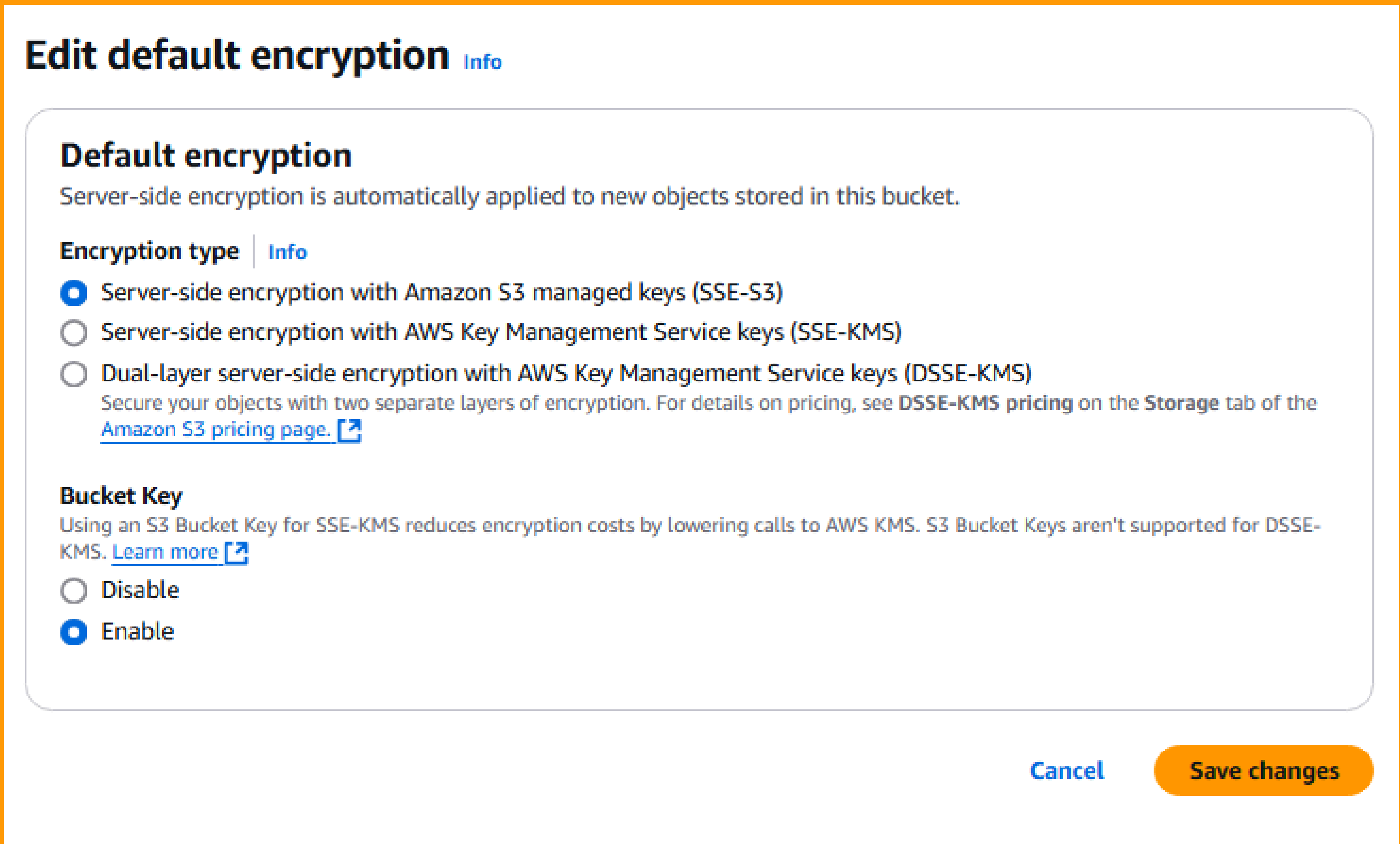
{} iam-policy.json > ...

```json
{
  "Version": "2012-10-17",
  "Statement": [
      {
          "Sid": "AccessSourceBucket",
          "Effect": "Allow",
          "Action": [
              "s3:ListBucket",
              "s3:GetObject"
          ],
          "Resource": "arn:aws:s3:::<BucketName>/*"
      }
  ]
}
```

S3 Data Encryption

Protect your data with encryption at rest and in transit.

- **Server-side encryption (SSE)**: SSE-S3, SSE-KMS, SSE-C.
- **Client-side encryption**: Data encrypted before uploading to S3.



Access Management

Control who can access your data with flexible access policies.

- AWS Identity and Access Management (IAM).
- Bucket policies and Access Control Lists (ACLs).
- AWS Organizations for cross-account access.

Access Management
Access Control List

Access control list (ACL)

Edit

Grant basic read/write permissions to other AWS accounts. Learn more

Public access is blocked because Block Public Access settings are turned on for this bucket
To determine which settings are turned on, check your Block Public Access settings for this bucket.
Learn more about using Amazon S3 Block Public Access

The console displays combined access grants for duplicate grantees
To see the full list of ACLs, use the Amazon S3 REST API, AWS CLI, or AWS SDKs.

Grantee	Objects	Bucket ACL
Bucket owner (your AWS account) Canonical ID: 744c069beddb1e2afa376ee94c5260f2c5358cd14497488c4c1729f2c870483a	List, Write	Read, Write
Everyone (public access) Group: http://acs.amazonaws.com/groups/global/AllUsers	-	-
Authenticated users group (anyone with an AWS account) Group: http://acs.amazonaws.com/groups/global/AuthenticatedUsers	-	-
S3 log delivery group Group: http://acs.amazonaws.com/groups/s3/LogDelivery	-	-

Access Management
Public Access

Object URL

https://my-aws-bucket-unique.s3.eu-central-1.amazonaws.com/AWS-logo-s3.png

This XML file does not appear to have any style information associated with it. The document tree is shown below.

▼<Error>
<Code>AccessDenied</Code>
<Message>Access Denied</Message>
<RequestId>304PT6C9Y4QB3SDN</RequestId>
<HostId>4WMQy/S9Jc7XL0ht7wii+w7pKbu0OGDdqkJDYTU6/eaQ+cbiWsRV0X1M14v0m5QccDmwqZDtcSc=</HostId>
</Error>

Edit Block public access (bucket settings) Info

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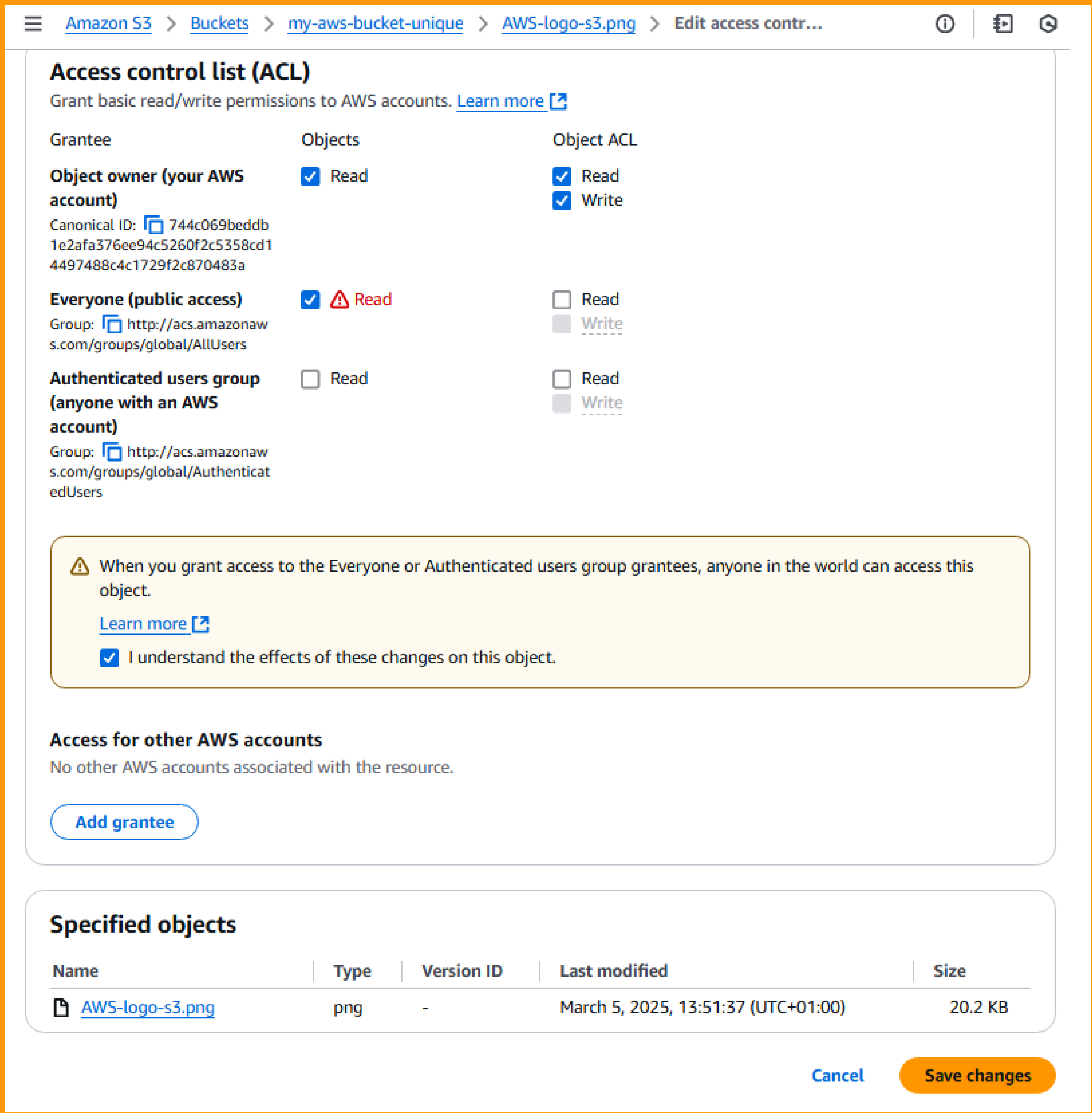
Cancel | Save changes

Access Management
Public Access



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Access Management
Public Access

Object URL

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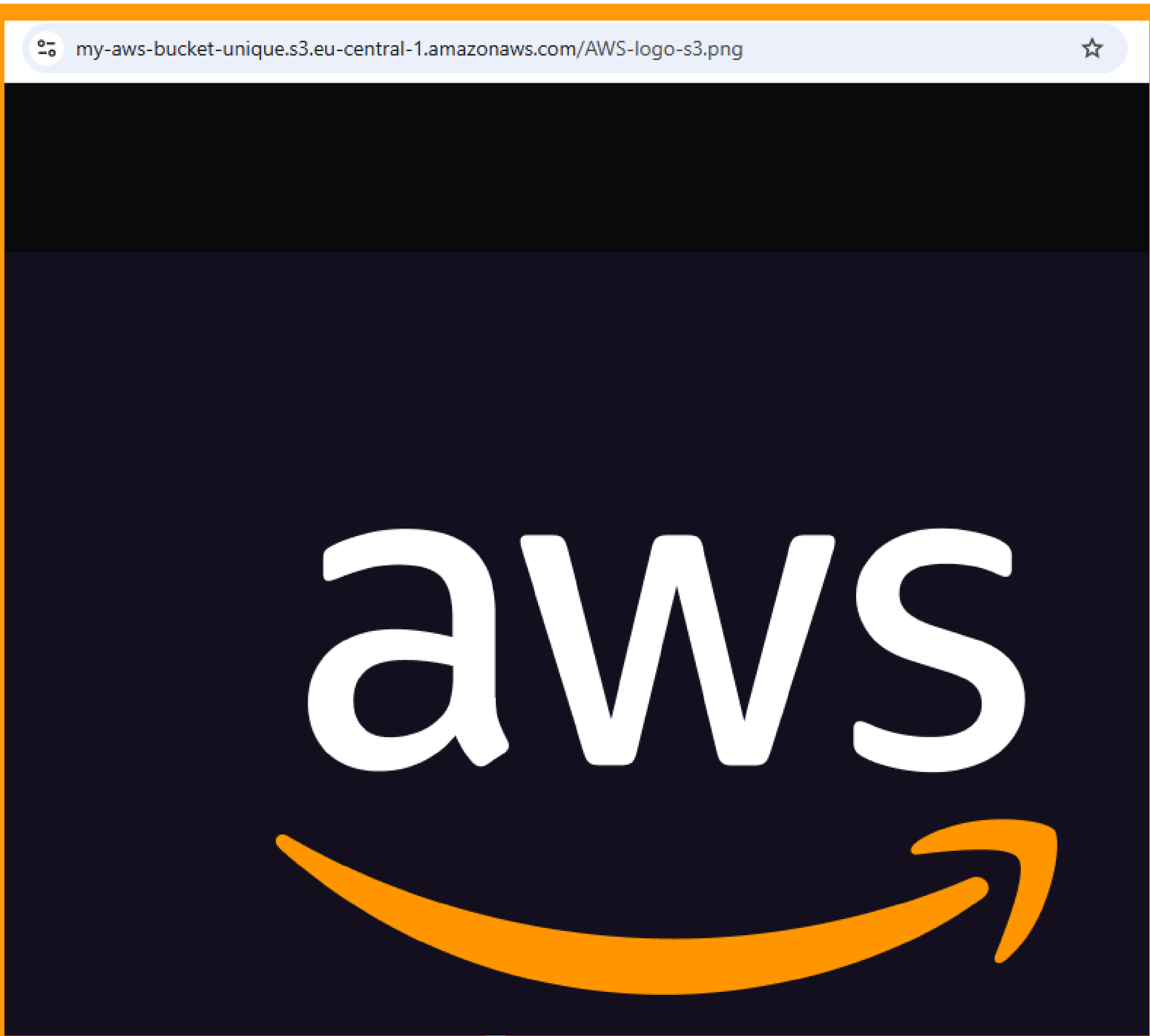
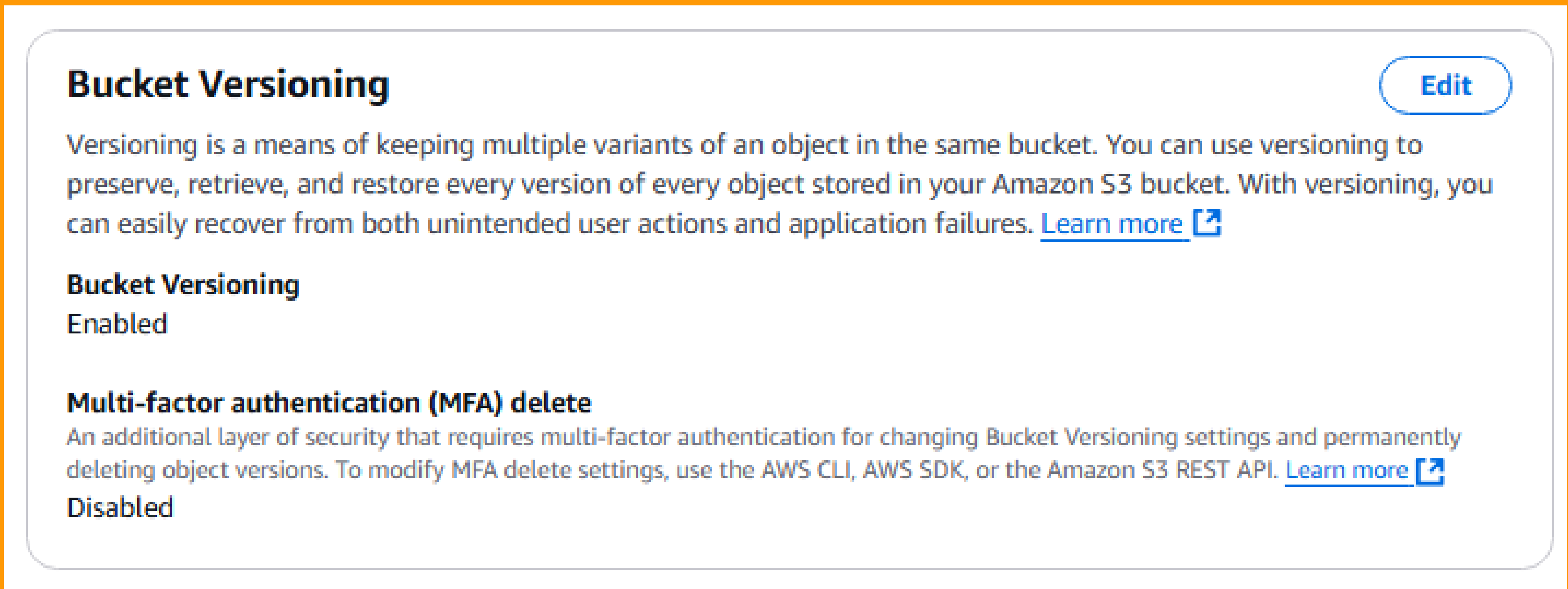


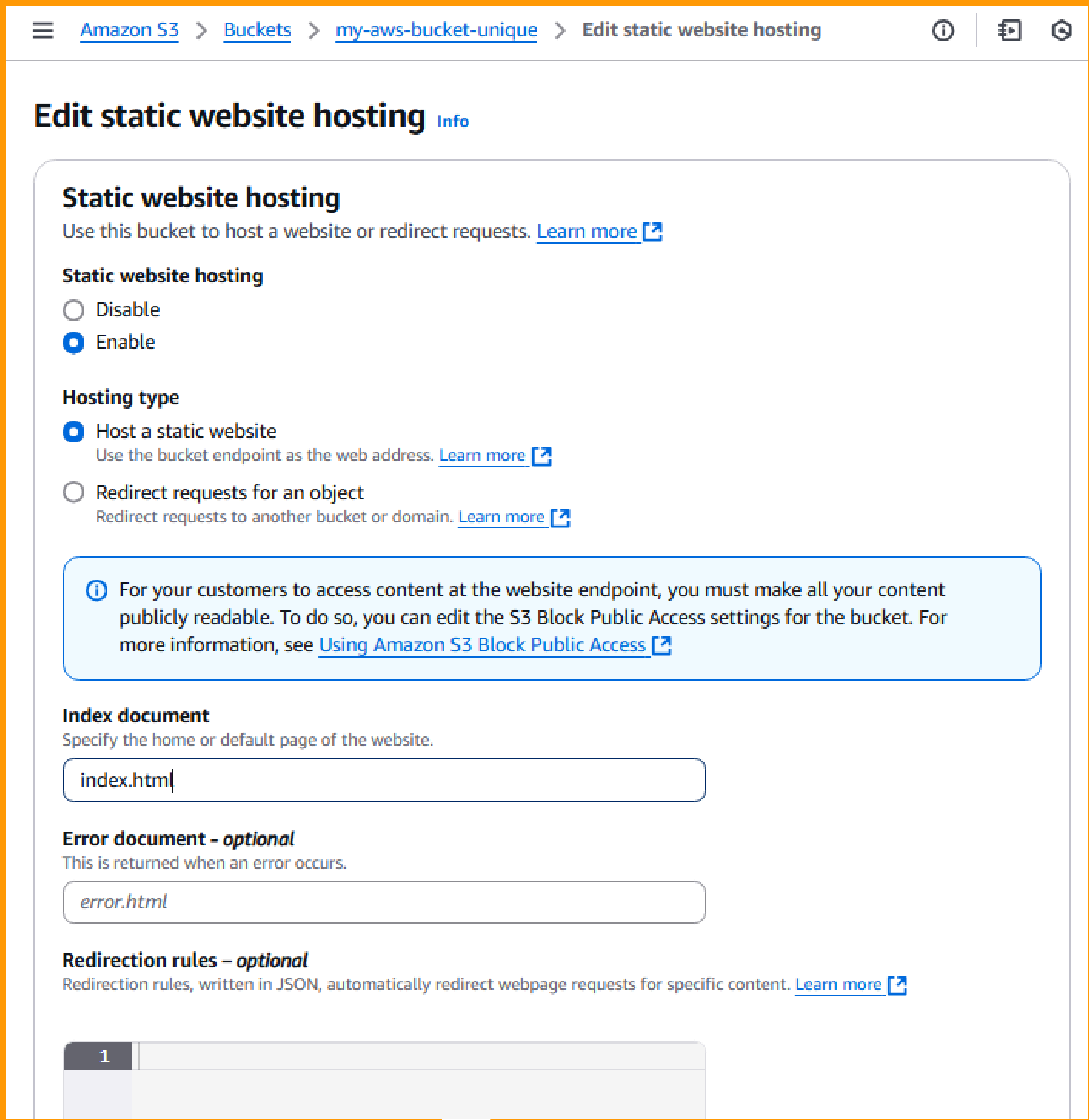
S3 Object Lock and Versioning

- **Object Lock**: Prevents data deletion or modification.
- **Versioning**: Maintains multiple versions of an object.

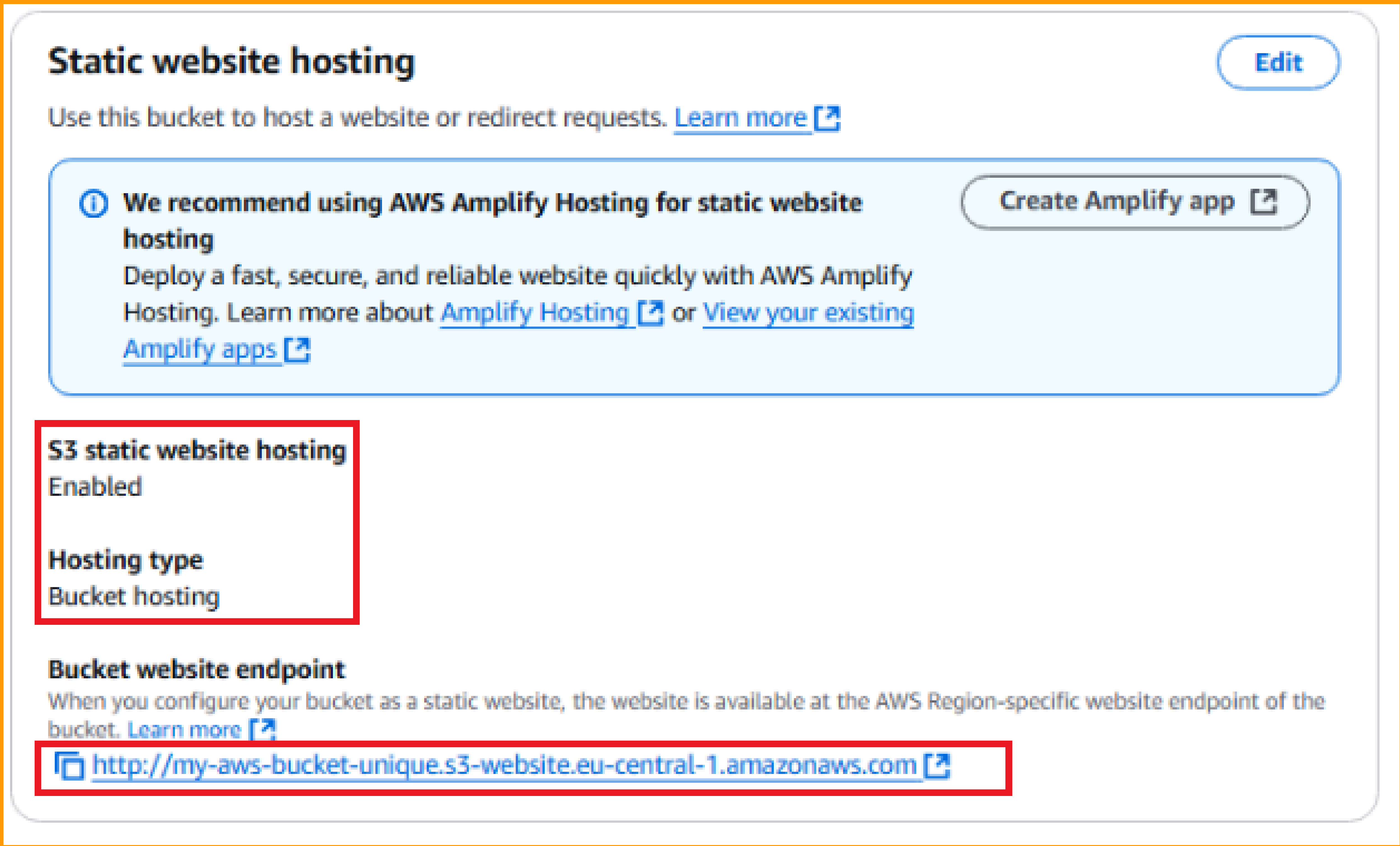


Static Website Hosting

- Host static websites directly from an S3 bucket.
- Configure bucket permissions for public access



Static Website Hosting



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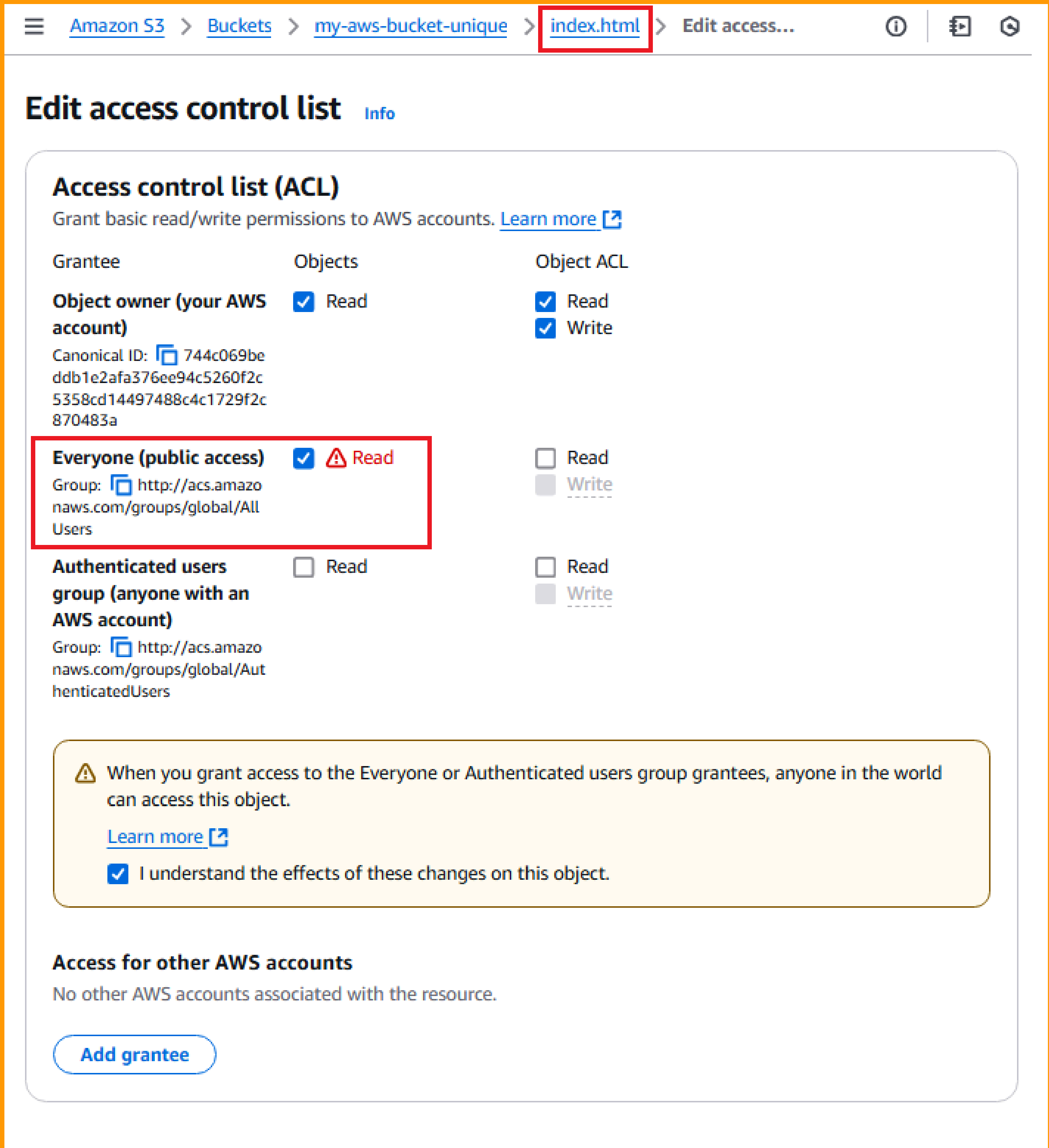

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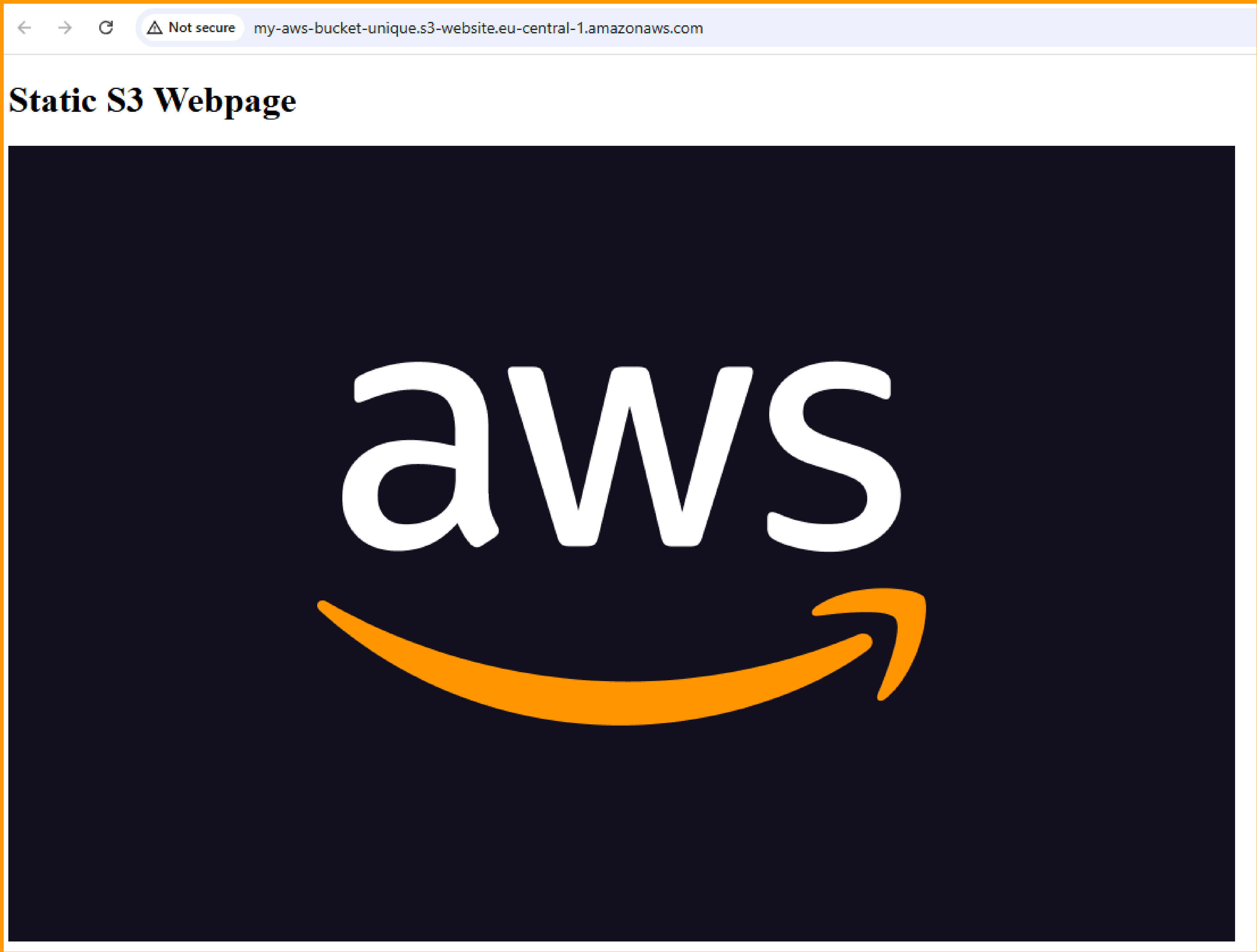
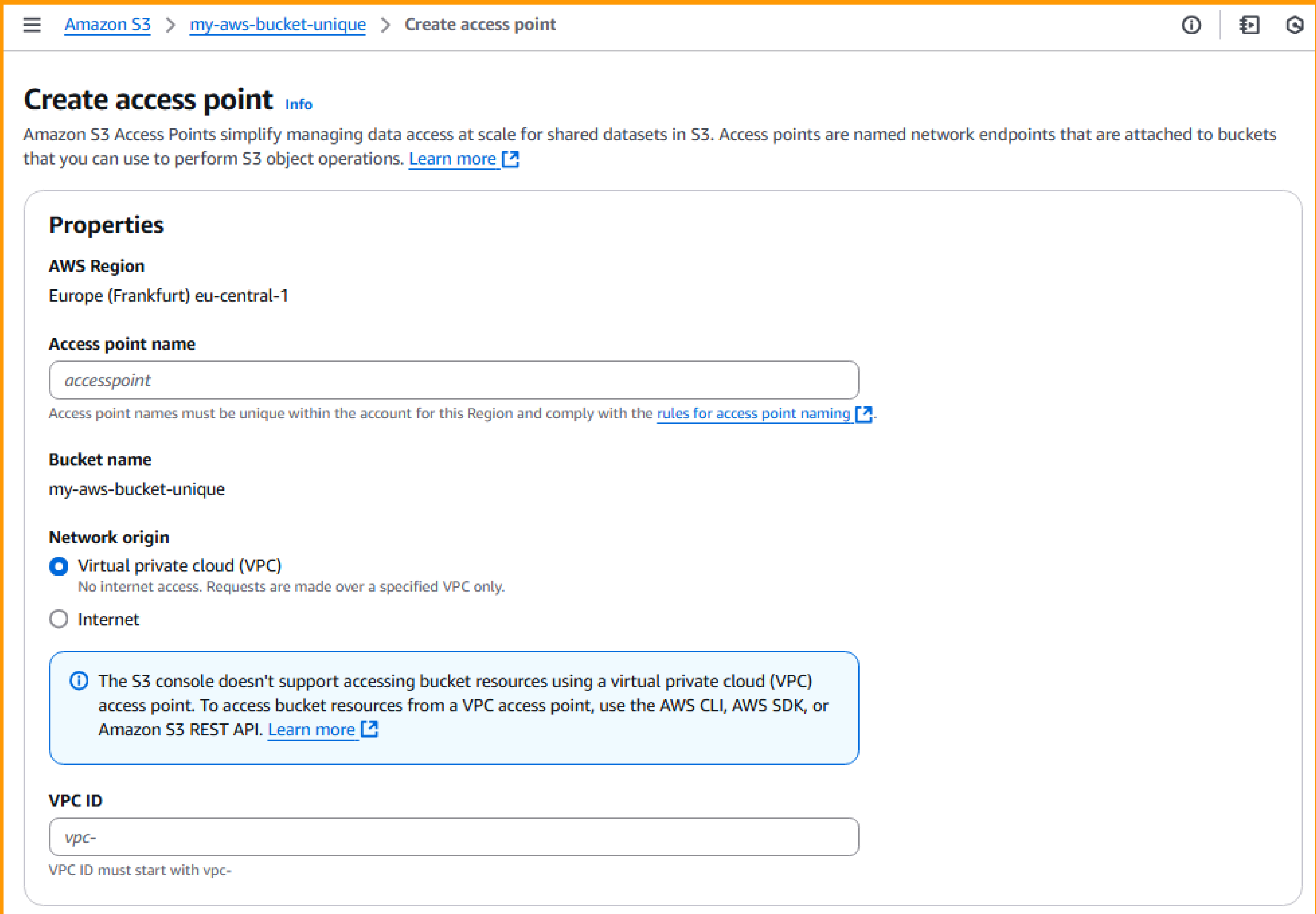


S3 Access Points

S3 Access Point is a feature that provides a **dedicated, customizable endpoint** to simplify and **control access** to **S3 buckets** for specific applications or user groups



```
user@aws:$  ####################################
```

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Feel free to like, share, or repost
to help more people see it

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